Weekly report

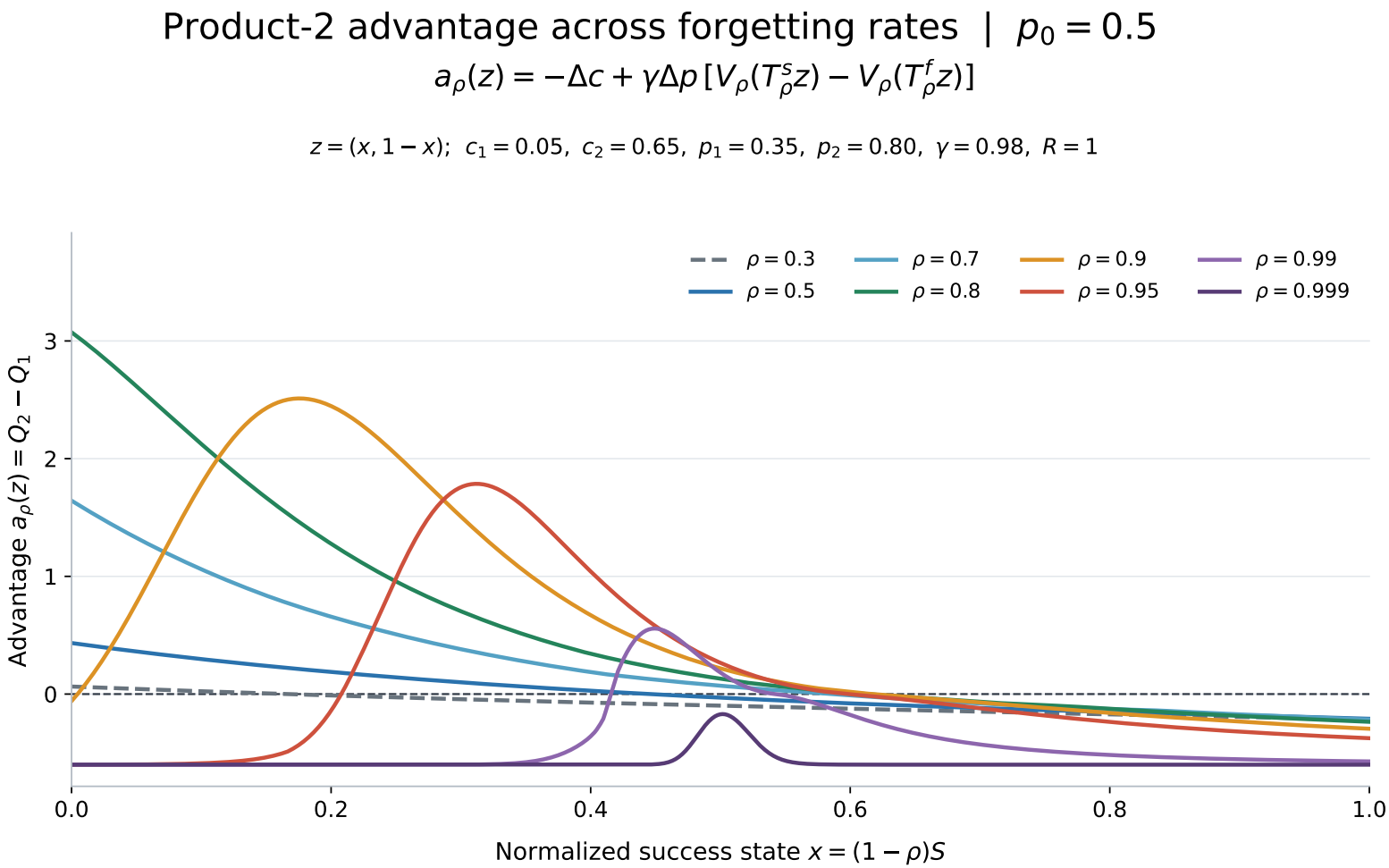
Natalia Trigo Tomasevich

September 29, 2026

From last week

- Try to prove that the solution was a threshold policy
- The fluid model

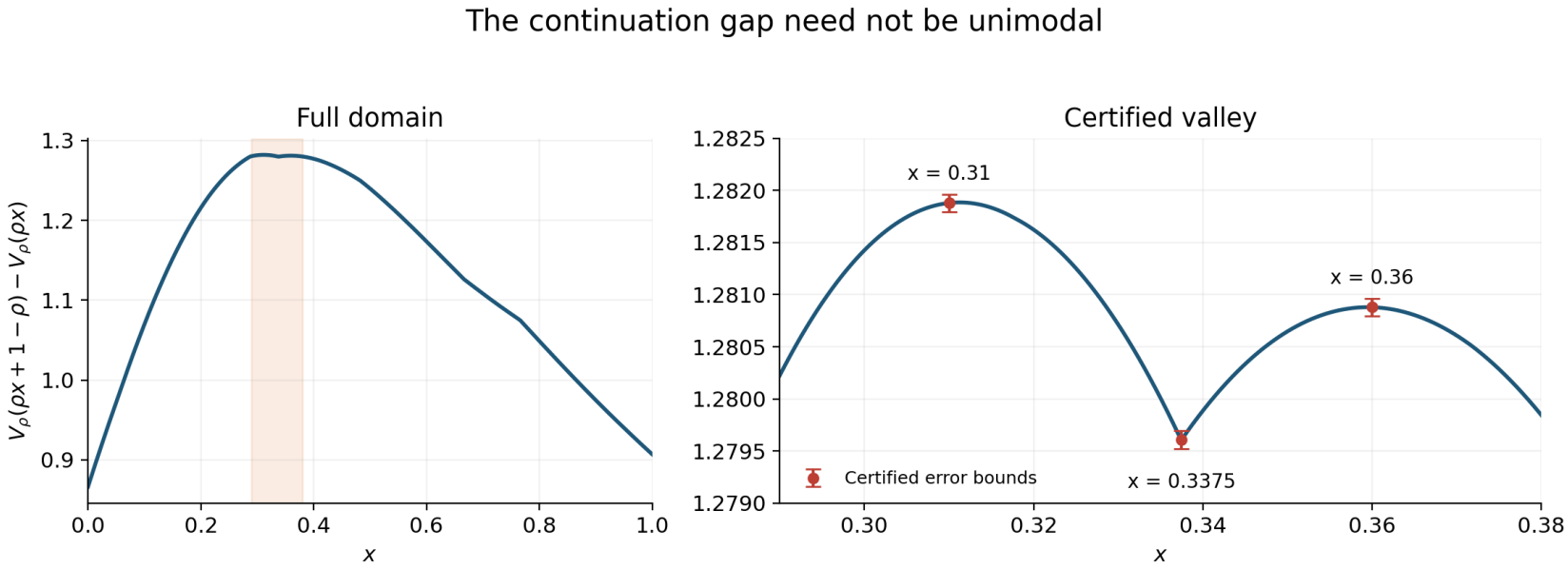
The base case



Above the dashed zero line, product 2 is preferred; below it, product 1 is preferred.

Curves cover the full mature boundary. For $\rho < 0.5$, only a Cantor subset is limiting-reachable.

No unimodality



$\rho = 0.7,\ p_0 = 0.7,\ p_1 = 0.1,\ p_2 = 0.9,\ c_1 = 0,\ c_2 = 0.925,\ R = 1,\ \gamma = 0.95$

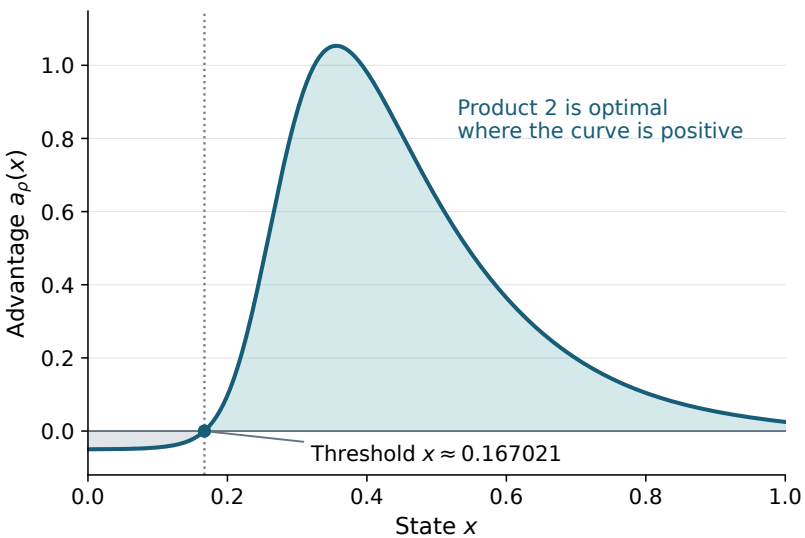
A weird case

One lower threshold: the policy is $1 \to 2$

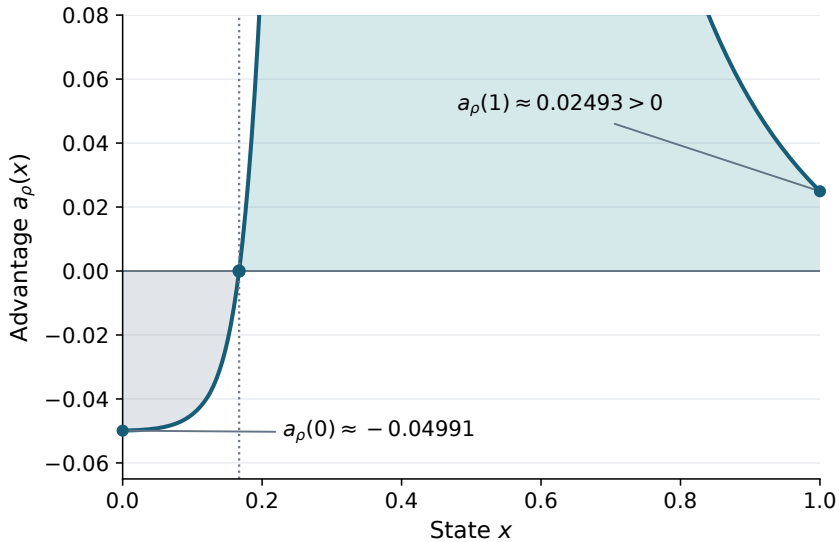
$\rho = 0.95,\ \Delta p = 0.20,\ \Delta c = 0.05$ | $p_1 = 0.35,\ p_2 = 0.55,\ c_1 = 0.05,\ c_2 = 0.10$

$R = 1,\ \gamma = 0.98,\ p_0 = 0.50$

Full advantage curve



Vertical zoom around zero

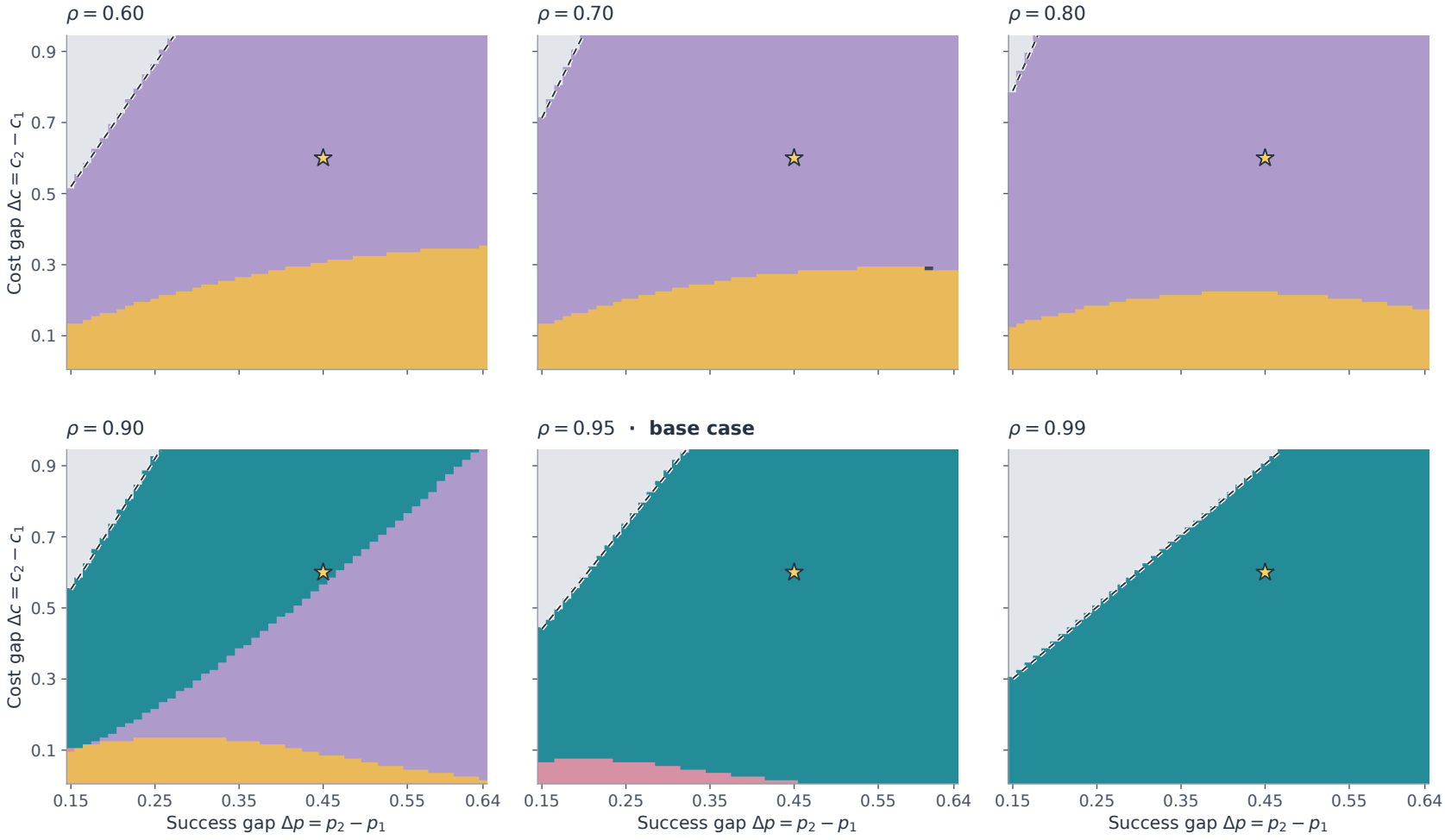


Product 1: $0 \le x < 0.167021$ Product 2: $0.167021 < x \le 1$ Both products tie at the threshold.

$a_\rho(x) = -\Delta c + \gamma \Delta p[V_\rho(\rho x + 1 - \rho) - V_\rho(\rho x)]$ | Numerical result; checked at 8,001 and 16,001 state nodes.

Where does the two-threshold policy appear?

$R = 1$, $\gamma = 0.98$, $c_1 = 0.05$, $p_1 = 0.35$, $p_0 = 0.50$ | Product sequence is read as state x increases.



Each tile represents a solved parameter pair; finite-grid numerical evidence. The domain satisfies $p_0 \le p_2$.

About the thresholds

- We don't always have thresholds. It depends on the parameters of the problem
- For some parameters we do have thresholds
- We could have up to 2 threholds.
- They are always the same patterns

$$1 \to 2, \qquad 1 \to 2 \to 1, \qquad 2 \to 1$$

We don't always have thresholds

If we have

$$\mathcal{A}^1_\rho = \max_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[U^1_\rho(T^s_\rho z) - U^1_\rho(T^f_\rho z) \right] \right\} < 0$$

Then for all z we have that to use product 1 is optimal.
If we have

$$\mathcal{A}^2_\rho = \min_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[U^2_\rho(T^s_\rho z) - U^2_\rho(T^f_\rho z) \right] \right\} > 0$$

Then for all z we have that to use product 2 is optimal.

We have thresholds

We have

$$\mathcal{A}^1_\rho = \max_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[U^1_\rho(T^s_\rho z) - U^1_\rho(T^f_\rho z) \right] \right\} > 0$$

$$\Longleftrightarrow$$

$$\mathcal{A}_\rho = \max_{z \in \mathcal{K}_\rho} \underbrace{\left\{ -\Delta c + \gamma \Delta p \left[V_\rho(T^s_\rho z) - V_\rho(T^f_\rho z) \right] \right\}}_{a_\rho(z)} > 0$$

We have thresholds

I proved that

$$\mathcal{A}^2_\rho = \min_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[U^2_\rho(T^s_\rho z) - U^2_\rho(T^f_\rho z) \right] \right\} < 0$$

$$\Longleftrightarrow$$

$$\mathcal{A}'_\rho = \min_{z \in \mathcal{K}_\rho} \left\{ -\Delta c + \gamma \Delta p \left[V_\rho(T^s_\rho z) - V_\rho(T^f_\rho z) \right] \right\} < 0$$

Threshold area

If $\mathcal{A}_\rho^1 > 0$ and $\mathcal{A}_\rho^2 < 0$ we have at least one threshold,because that condition means that $\exists z_1, z_2 \in \mathcal{K}_\rho$ such that $a_\rho(z_2) < 0 < a_\rho(z_1)$. Which means, by continuity and intermediate value that $\exists z^* \in (min\{z_1, z_2\}, max\{z_1, z_2\})$ such that $a_\rho(z^*) = 0$.

Conditions over c_2

- If we fix all the parameters except for c_2, with the conditions $\mathcal{A}_\rho^1 > 0$ and $\mathcal{A}_\rho^2 < 0$ we can find an upper bound and lower bound for c_2 where we can find thresholds.
- The proof is not revised yet.

The fluid model

Proceed to progress note...